CS 353 - Analysis of Algorithms
Spring 2026
Practice Exam 2
Due 3/9/26

Name: gwynnie hayes

I pledge my honor that I have neither given nor received assistance during this exam, and that I have seen no dishonest work.

Signed _____

☐ I have intentionally not signed the pledge (check only if appropriate)

Standard ORDER. Directly from the definition, prove that a given function is order another function, and a given function is not order another. List given functions by increasing order of growth, and describe the complexity class of each function.

1. Prove that $3n^2 + 2$ is order n^3.

we'll show that there exist positive integers such c and n_0 such that if $n \geq n_0$, then $3n^2+2 \leq cn^2$.

let $c = 5$ and $n_0 = 1$.

$$\begin{aligned} 3n^2 + 2 &\leq 3n^2 \cdot n + 2 && \text{because } n \geq 1 \\ &= 3n^3 + 2n^3 && \text{because } n \geq 1 \\ &= 5n^3 \\ &= cn^3 && \text{because } c = 5 \end{aligned}$$

Thus if $n \geq 1$ then $3n^2+2 \leq cn^3$ which shows that $3n^2+2$ is order n^3.

2. Prove that $4n^3 + 2$ is not order n.

for any positive integers c and n_0
there exist some integer n with $n \geq n_0$
and $4n^3+2 > cn$.

For c and n_0 let $n = \max\{n_0, c+1\}$

$$
\begin{aligned}
4n^3+2 &= (4n^2 \cdot n)+2 \\
&> 1 \cdot 4n^2+2 \quad \text{because } n \geq c+1 > 1 \\
&= 4n \cdot n+2 \\
&\geq 4n(c+1)+2 \quad \text{because } n \geq c+1 \\
&= 4nc+4n+2 \\
&> cn \quad \text{because } n \geq c+1 > 1
\end{aligned}
$$

Thus $4n^3+2 > cn$, so $4n^3+2$ is not order n

3. List the following functions in increasing order of growth (from slowest growth to fastest growth), and indicate the complexity class of each function.

   (a) $2^n + 3^n$
   (b) $2n\log(n) + 5\log(n)$
   (c) $4 + 5n + 3n^3$
   (d) $n^3 + n\log(n)$
   (e) $\log(n) + n + 2^n$
   (f) $43\log(n) + 2$
   (g) $3n^2 - n + 5$
   (h) $n\log(n) + 3n$

- $43 \log(n) + 2$
- $2n\log(n) + 5\log(n)$
- $n\log(n) + 3n$
- $3n^2 - n + 5$
- $n^3 + n\log(n)$
- $4 + 5n + 3n^3$
- $\log(n) + n + 2^n$
- $2^n + 3^n$

☐ **P**: You have demonstrated proficiency in this standard and have earned full credit, congratulations!
☐ **R**: You are close to proficiency, but there are some issues with your solution. If you submit a correct revision by 3/23, this can be changed to a P.
☐ **S**: You are starting to show partial proficiency in this standard, and have earned half credit. You will be able to schedule a second attempt on this standard.
☐ **I**: You have not yet demonstrated proficiency in this standard, but you will be able to schedule a second attempt on this standard.